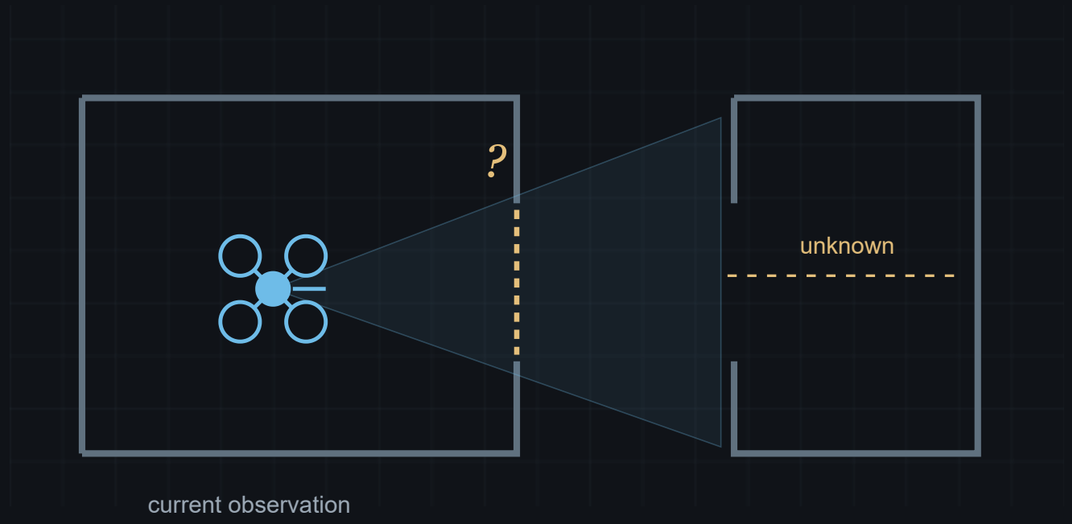
The course is visible.
The next action is uncertain.

Indoor navigation, incomplete observations, and a mission beyond the field of view.



What should happen next?

A passage candidate

What lies beyond it?

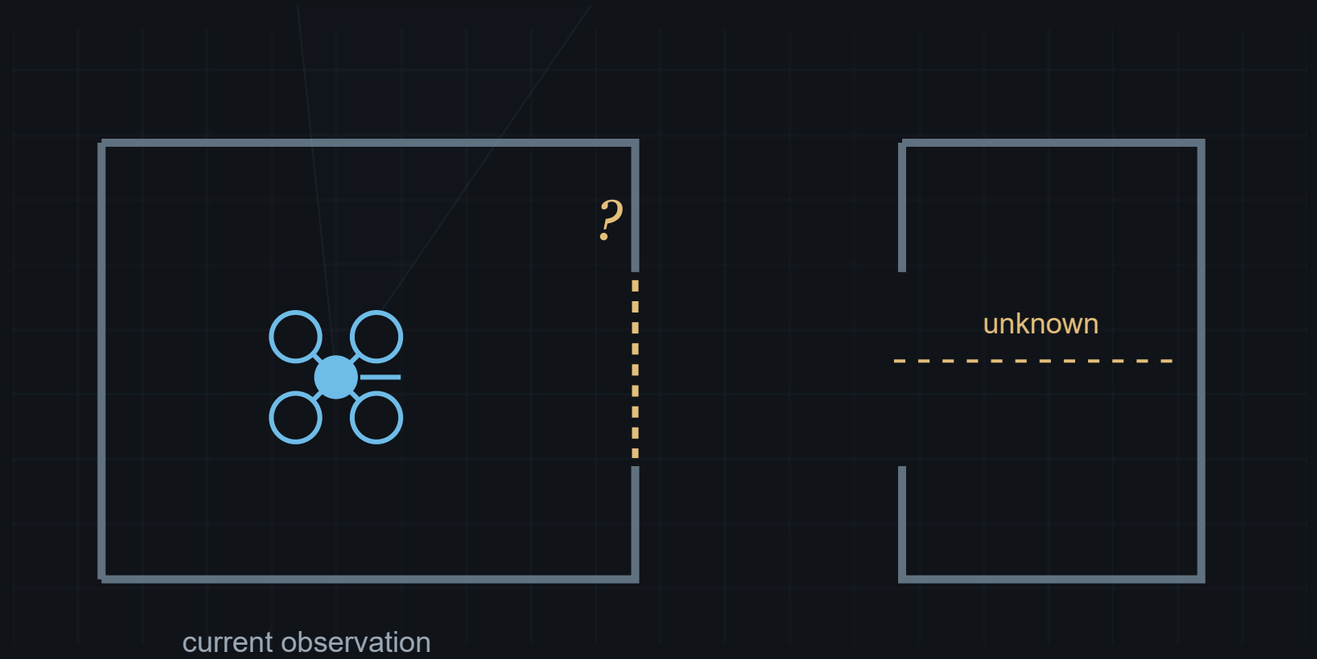
Observe before committing.

geometry + memory + intent

↳ Each camera frame reveals only part of the task.

Build what the drone *remembers*.

Remember openings, keep uncertainty, and look again when needed.



Memory ≠ a fresh view

opening_0003

same entity, new viewpoint

last seen → aging

identity persists · certainty does not

↳ Prioritize a useful world representation before increasing policy complexity.

Illustrative belief update

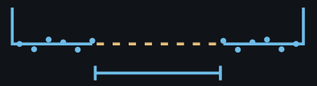
Give the map meaning.

Research gave us the structure; our constraints shaped the implementation.

RESEARCH → ADAPTATION

Hydra

build the map as we move



CSIRO: recovery in partial observation

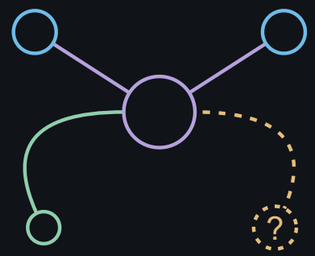
Kimera

connect geometry to meaning



ModalAI: separate perception, planning and autopilot interfaces

A useful world memory



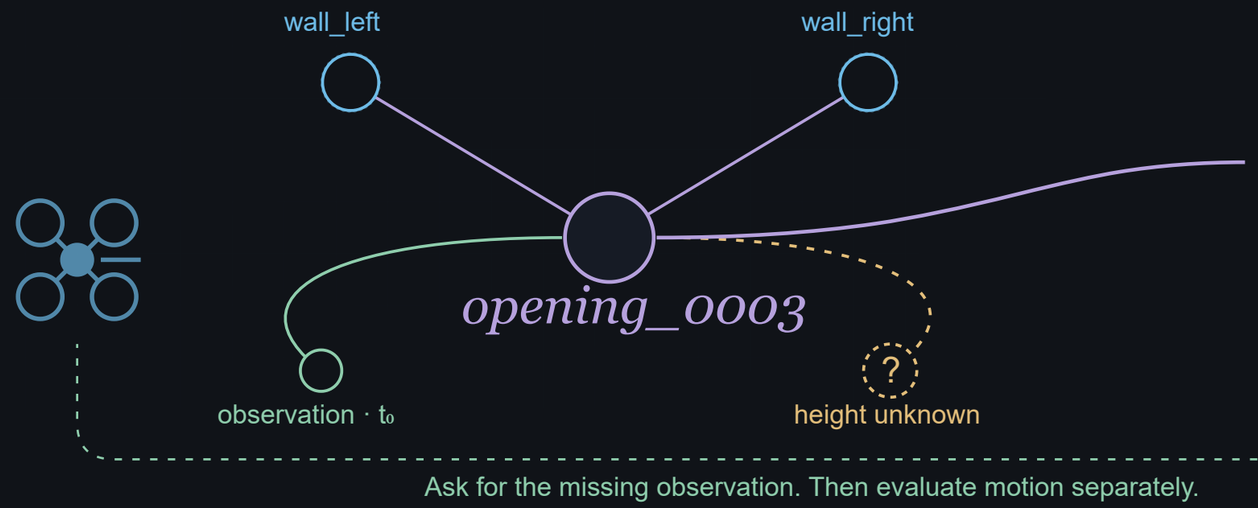
geometry + idopening + observations

↳ A compact adaptation for our task and compute budget.

From seeing to *understanding*.

Observations become geometry, then meaning, then a grounded decision.

01 OBSERVATIONS | 02 GEOMETRY | 03 SEMANTIC GRAPH | 04 REASONING



GOAL PROPOSAL

observe_more

target: opening_0003

because: height is unknown

Grounded proposal ≠ permission to fly

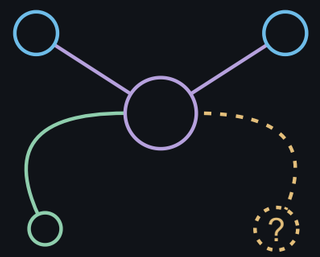
Ask for the missing observation. Then evaluate motion separately.

↳ Missing evidence becomes a reason to look again.

Illustrative entity · offline contract

Sometimes the next action is *to look*.

If we do not know whether the drone fits, take another look.



opening_0003

height remains unknown

Read the memory

opening + observations

what we know + what is missing

do not assume a way through

Choose a useful action

observe_more

target: opening_0003

because: height is unknown

Check the evidence is current

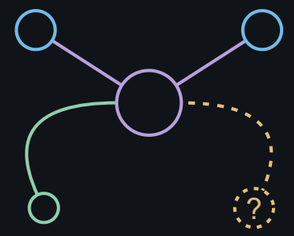
Motion still needs its own feasibility check.

↳ When important information is missing, look again before committing.

Non-executable proposal

Remember. Decide. Fly.

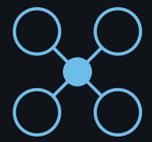
Three jobs, with a safety check before movement.











1. Remember

What is here?

2. Decide

What should we do next?

3. Fly safely

Can we make this move?



New observations update memory

↳ Reasoning chooses the goal. Control turns it into movement.

Why DATT and physics-informed learning?

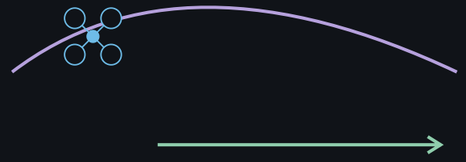
A good route still needs a drone that can follow it when conditions change.



Planned ≠ actual

wind · delay · model error

DATT · follow the route



Look ahead + adapt

Correct for disturbances

Proposed tracking experiment

PINNs · predict motion

Measurements

+ physical equations

A physics-guided prediction

Learn without ignoring physics

Our predictor: offline-tested

↳ Use learning to handle motion mismatch; keep world understanding separate.

Our Classical baseline: plan, then follow.

Established algorithms give us a clear starting point to measure against.

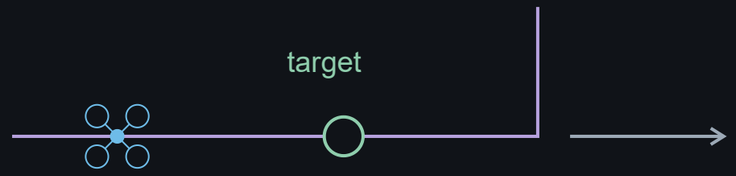
Plan a clear route



Dijkstra grid search

Live planner: A* alternative

Follow a sliding target



Look ahead; wait at corners

Limit lead to avoid cutting turns

Send position targets



PX4 flight control

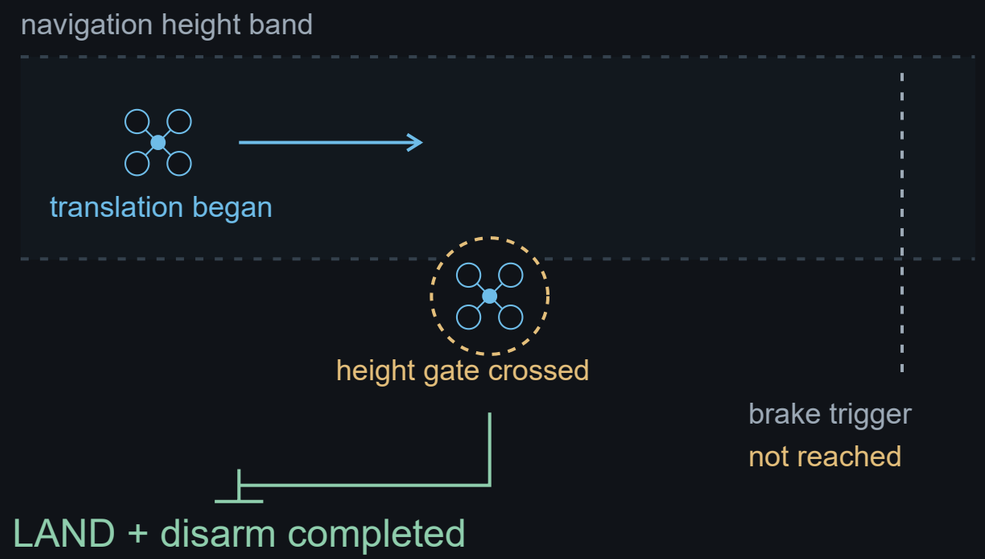
Monitor height and contact

↳ An understandable baseline lets us measure whether learning adds value.

What failed — and what we changed.

Simulation exposed problems with height, coordinates and timing.

HEIGHT DROPPED BEFORE THE BRAKING TEST



Schematic states and event order · not a measured trajectory

FOUR ATTEMPTS COULD NOT START



01 Coordinates disagreed

use the same position reference



02 Clock alignment

preserve capture-time identity



03–04 Data arrived too slowly

keep the command loop responsive

No course-flight success claimed.

↳ Fix height, shared coordinates and timing before adding more intelligence.

What we built. What we still need to prove.

Evidence is specific to a component, an experiment and an environment.

EVIDENCE IS LOCAL TO THE CLAIM



Offline components

entities · reasoning · mapping

synthetic comparison evidence



Simulation checks

Hover check: completed

Braking test: landed; test failed



Mission-level benefit

hybrid course + matched trials

hardware qualification remains

Separate baseline: Classical S1 has one successful Gazebo course run.

↳ The hybrid architecture has not yet demonstrated full-course native success in the fetched record.

From observations to *justified actions*.

The next milestone is a measurable mission benefit from each layer.



01 Fly one reliable segment

coordinates · timing · height



02 Matched comparison

same map, start and scoring



03 Layer ablations

memory → topology → semantics

Observations ⟶ *Grounded belief* ⟶ *Justified actions*

Learned extensions only when the evidence warrants them.

↳ Prove reliable flight, then measure what memory and reasoning add.

A map can say where. Semantics asks what next.

Three representations

- **Geometry:** surfaces, pose, observed free space and clearance.
- **Topology:** places and supported connections.
- **Semantics:** task-relevant meaning, such as a candidate passage.

Why keep memory?

Retain identity when a feature leaves view. Track stale or contradictory evidence. Select an information-gathering goal rather than inventing a connection.
These are architectural capabilities; mission benefit still needs a matched ablation.

Learning is a replaceable component.

Reasoning comparison

- G0: deterministic baseline.
- G1: utility / information-gain selection.
- G2: compact learned candidate ranking, evaluated with three seeds.

Motion prediction

D0 is the nominal physics default. Residual and adaptive alternatives exist offline, but clean-case improvements did not satisfy every delay, heading and runtime gate.
A learned predictor is not a learned navigation policy or a VLA controlling the aircraft.

Compare the same task, under the same conditions.

Reasoning benchmark

60 held-out layouts × 7 regimes × 6 steps. 420 episodes per variant.
Evidence acquired: G0 68; G1 260; G2 259 / 244 / 243.
Abstract synthetic evidence acquisition, not course completion.

Separate Classical S1 result

One Gazebo run: four checkpoints, pad landing and disarm in 84.3 seconds; zero touches reported.
Run `run-AdMalPxA`, revision `ad31ae5`. Not a matched hybrid comparison or physical-hardware result.

A useful fact needs a frame and a clock.

Data contracts

- Capture time, frame, units and reset epoch.
- Stable identity and supporting observations.
- Bounded context and independent validation.
- One command owner.

Four separate layers

- Permitted reference geometry.
- Persistent current-run belief.
- Fresh local collision evidence.
- Semantic and topological state.

A prior map does not become a fresh sensor observation.

A recovered experiment can still be a failed test.

Preserved outcomes

C1 crossed its height threshold before the brake trigger. LAND and disarm completed, but no braking envelope was measured. Latest fetched S0 attempts 001–004 stayed disarmed. Frame, timestamp and scheduling work remained.

Acceptance sequence

Native segment → matched baseline → memory / topology / semantics ablations → changed-layout evaluation.
Measure mission progress, false-feasible choices, observation cost and latency. Add learned complexity when these improve.

The ideas we adapted.

Academic references

Hydra · Hughes et al. · RSS 2022
Incremental hierarchical scene graphs.

Kimera · Rosinol et al. · 2021
Metric-semantic spatial representation.

CSIRO Data61 · Hudson et al. · 2021
Fielded subterranean robotic systems.

Engineering and visual references

ModalAI VOXL Vision Hub
VIO, obstacle processing and autopilot interfaces.

3Blue1Brown / Manim
Explanatory animation inspiration. All diagrams here are original SVG illustrations.

Methods inform our design; their published results do not validate our platform.